



# THE NINJA RUSH

A “CHROME DINO GAME” LIKE GAME

## MEMBERS OF OUR GROUP:

| NAME          | REGISTRATION NO.  |
|---------------|-------------------|
| MUHAMMAD ADIL | 24-NTU-CS-FL-1054 |
| MUHAMMAD ASAD | 24-NTU-CS-FL-1058 |
| ABDUL AHAD    | 24-NTU-CS-FL-1051 |

## SUBMITTED TO:

DR. SALMAN RAZA

## DATE OF SUBMISSION:

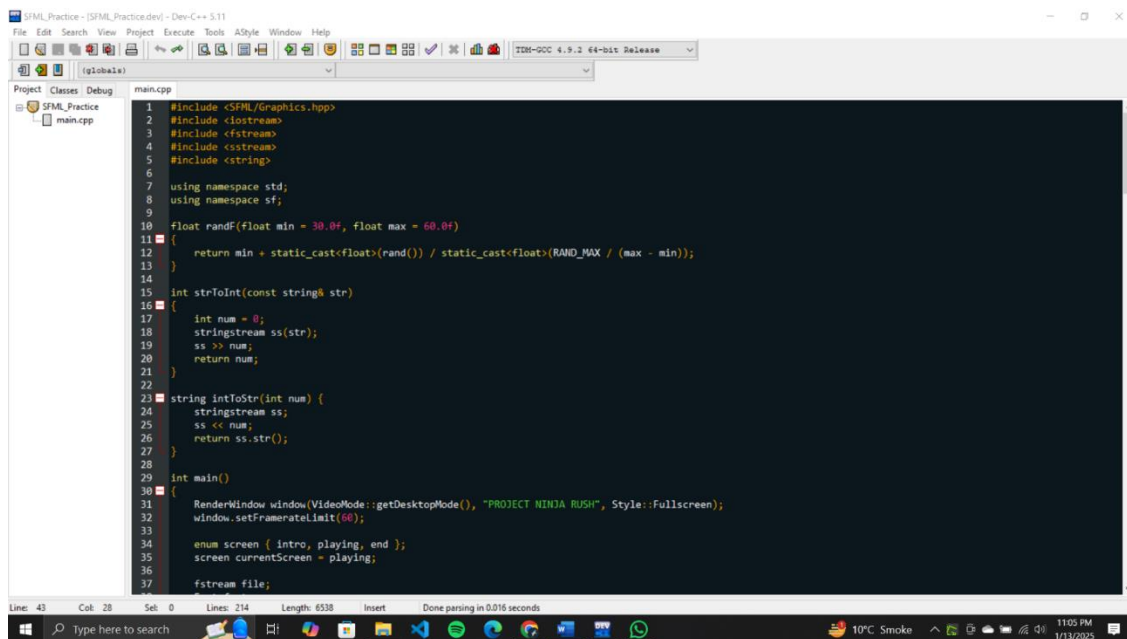
14/01/2025

## OUTLINES:

|                             |   |
|-----------------------------|---|
| CODE OVERVIEW.....          | 2 |
| PROBLEM STATEMENT.....      | 3 |
| RESEARCH OBJECTIVE.....     | 3 |
| METHODOLOGY.....            | 3 |
| DIFFICULTIES IN MAKING..... | 4 |
| USAGE OF THIS GAME.....     | 4 |

## CODE OVERVIEW:

### Input:



```
1 #include <SFML/Graphics.hpp>
2 #include <iostream>
3 #include <fstream>
4 #include <sstream>
5 #include <string>
6
7 using namespace std;
8 using namespace sf;
9
10 float randF(float min = 30.0f, float max = 60.0f)
11 {
12     return min + static_cast<float>(rand()) / static_cast<float>(RAND_MAX / (max - min));
13 }
14
15 int strToInt(const string& str)
16 {
17     int num = 0;
18     stringstream ss(str);
19     ss >> num;
20     return num;
21 }
22
23 string intToStr(int num) {
24     stringstream ss;
25     ss << num;
26     return ss.str();
27 }
28
29 int main()
30 {
31     RenderWindow window(VideoMode::getDesktopMode(), "PROJECT NINJA RUSH", Style::Fullscreen);
32     window.setFramerateLimit(60);
33
34     enum screen { intro, playing, end };
35     screen currentScreen = playing;
36
37     fstream file;
```

**Output:**



**PROBLEM STATEMENT:**

People in low-connectivity or offline situations have a shortage of interesting activities to do. That is why they often get bored and unproductive. The **Chrome Dino** game acts as a great offline purpose to kill time. A **C++ version** or some alteration of a game like this could also act as a means of entertaining people while learning programming, designing and problem-solving as well.

**RESEARCH OBJECTIVE:**

The aim of this report is to come up with a simple yet engaging Chrome Dino game using C++ programming language. The objectives are as follows:

1. The usage of C++ concepts such as classes and their fundamental concepts like inheritance and file handling would be put to practice.
2. The use of obstacle movement combining the ability of collision detection, scoring, and handling user input would be able to be implemented .

3. The use of basic graphics libraries for the creation of a simple display would be implemented.
4. The objective of promoting the use of C++ for game design in offline settings would be achieved.

### **METHODOLOGY:**

1. **IDE** (Integrated Development Environment):

DEV C++

2. **COMPILER:**

GCC (GNU Compiler Collection)

3. **TOOLS:**

- A C++ library is used, called **SFML**(Simple and Fast Multimedia Library).
- **Header Files:**  
#include <SFML/Graphics.hpp>  
#include <string>  
#include <fstream>  
#include <sstream>

### **DIFFICULTIES IN MAKING THIS GAME:**

1. **Collision Detection:**

Fine-tuning collision boxes for accurate detection between ninja and obstacles.

2. **Text Alignment:**

Properly positioning text elements like score, name input, and game over messages.

3. **Jump Mechanism:**

Adjusting jump height and smooth falling for realistic movement.

4. **Screen Management:**

Managing multiple screens (intro, playing, game over) effectively.

5. **Score Management:**

Storing and comparing scores correctly in a text file.

6. **Randomization:**

Randomizing obstacle sizes and colors without breaking gameplay balance.

**7. Optimization:**

Balancing frame rate and game speed for smooth performance.

**USAGE OF THIS GAME:**

The Ninja Rush game can be used for:

1. **Learning Tool:** Helps beginners understand C++ concepts like classes, inheritance, and file handling.
  2. **Game Development Practice:** Provides experience with collision detection, scoring systems, and graphics handling.
  3. **Entertainment:** A fun offline game for casual play.
  4. **Coding Competitions:** It can be used as a project in coding contests or hackathons.
  5. **Educational Demos:** Useful for teaching basic game mechanics and event handling in C++.
-